Search & Rescue Coordinated Intelligence Systems

The autonomous teamwork of ground and air assets in a mission-oriented environment

SRCIS

"Perfectly balanced... As all things should be."

Yavanni Ensley

Younghoon Cho

Jaylin "Pop" Ollivierre

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OVERVIEW

- Progress Matrix
  - Accomplished Tasks
    - Milestone 5 Plan
      - Demos

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M4

Progress Matrix

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M4 Progress Matrix

Completed Tasks

Task	Completion	Yav	Young	Pop	To Do
Drone Setup (Parrot)	100%	40%	60%	0%	
Unitree Go2 Setup	100%	40%	0%	60%	
Target Sharing	80%	40%	20%	20%	Clean up bugs
Multi-agent Coordination	75%	50%	15%	10%	Unitree GO2 / Parrot integration
Concurrent Map Sharing	10%	10%	0%	0%	Implement an algorithm we researched to fit the needs of what we want; Merge the maps
Target engagement algorithm	0%	0%	0%	0%	Dependent on other tasks. Make an efficient algorithm for multiple robots to surround the target.
Robot Operator	0%	0%	0%	0%	Dependent on other tasks; need to finish the target engagement algorithm

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M4 Progress Matrix

Completed Tasks

Parrot drone setup

- drone is able to
  - land/takeoff
  - operate
  - live stream

With safety code

Unitree G02 setup

- Fully remote operable
- Limo code reusable

Target sharing

- When target is recognized, other robots who don't have the target in sight is able to calculate the fastest route considering obstacles.

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M4 Progress Matrix

Completed Tasks

Multi-agent Coordination

- Unitree
- Parrot

Needs to be added

Concurrent Map Sharing

- working on different robots to be on same origin map

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M5

Task Matrix and Rollovers

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Target engagement algorithm

- Unitree
- Parrot

Needs to be added to the current mapsharing

Robot Operator

- Deciding the confidence level of the target
- continuously track the target when target is unfound

M4 Task Rollover

M5 Task Matrix

M5 Tasks

Task	Yav	Young	Pop	To-Do
Target Sharing	50%	25%	25%	Clean up bugs
Multi-agent Coordination	60%	25%	15%	Setup Unitree GO2 + operation
Drone integration (Parrot)	40%	60%	0%	Integrate w/ Base Driver
Unitree Go2 Integration	40%	0%	60%	Integrate w/ Base Driver
Concurrent Map Sharing	60%	20%	20%	Implement an algorithm we researched to fit the needs of what we want; Merge the maps
Target engagement algorithm	33%	33%	33%	Dependent on other tasks. Make an efficient algorithm for multiple robots to surround the target.
Robot Operator	33%	33%	33%	Dependent on other tasks; need to finish the target engagement algorithm
Mission Command Structure	33%	33%	33%	Draft a scheme of maneuver to explain the priorities pre-, during, and post-execution
Full-dress Demo + Poster	33%	33%	33%	Dependent on other tasks; Mock demo

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Target sharing

- takes long time for robot to calculate route
- Robots gather map info when no target is present

Multi-agent coordination

- Drone(parrot)
- Unitree Go2

- make code for Limos work for Unitree
- add skyview of drone for mapping

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Concurrent Map Sharing

- share map to Unitree & Drone

Mission Command Structure

-Plan for execution and coordination

Full-dress Demo + Poster

- make demo and get feedback from customer(Professor)

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Demos

-Target Sharing
-Go2 Remote Operation

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LIMO - Target Sharing

Unitree Go2 Remote Operation

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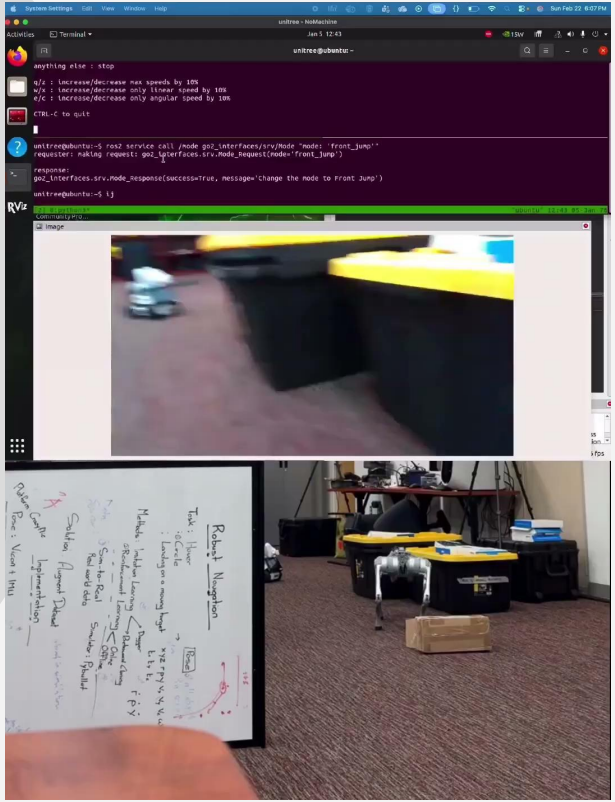
LIMO - Target Sharing

Unitree Go2 Remote Operation



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QUESTIONS?

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